Chapter 1: Probability Basic Concepts

NAT

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Basic Concepts of Statistics & Random Experiments

- Statistics involves collecting, classifying, summarizing, organizing, analyzing, and interpreting data
- A random experiment is an experiment where the outcome cannot be predicted with certainty.
- **Defining Characteristics of a Random Experiment:**
  (a) All possible outcomes of the experiment are known in advance.
  (b) Any single performance of the experiment results in an outcome that is not known in advance.
  (c) The experiment can be repeated under identical conditions.

Definition of a Set & Membership

- **Definition** A set is simply a collection of objects, these objects being called elements or members of the set.
- Examples include the set of all real numbers and the set of all UPNM students
- Membership of a set X by an object x is denoted by a special symbol $\in$, read as "belongs to"
- The statement $x \in X$ means that x is an element of, or belongs to, X .
- There is also an opposite statement $x \notin X$ which means that x is not an element of, or does not belong to, X .
- **Example:** For example, with S being the set of all students registered for TPQ3783, $\in S$ (enter your name)

Specifying sets

- Example of a set

$$S := \{1, 2, 3, 4, 5, 6\} \tag{1}$$

  - The notation ":= " means "is defined to be equal to
  - S is a set that contains exactly six numbers listed
  - The order of the elements does not matter. Its membership/ existance in the set is important
  - If the set is large (multiple/ many members/ elements) **and** have a clear pattern, ... is typically used. E.g.: $S := \{1, \ldots, 20\}$

Examples of Random Experiments

- **The experiment of tossing a coin:**
  - There are two possible outcomes: getting a Head (H) or getting a Tail (T).
  - Even though we know both **possible outcomes**, we cannot definitively predict that we will get a tail if we throw the coin in a particular manner.
- **The experiment of rolling a die:**
  - There are six possible outcomes: the die showing up 1, 2, 3, 4, 5, or 6.
  - We cannot for sure predict that we can make a particular number appear on the face of the dice when it is rolled.

Applied Real-Life Examples

- **Quality Control & Manufacturing:**
  - Testing the length of life of a 60-watt light bulb until it burns out.
    - There infinitely number of possible of outcomes. (Why?)
- **Environmental Testing:**
  - Counting the number of bacteria colonies in a 100-milliliter water sample taken from a specific lake basin.
- **Technology Reliability:**
  - Measuring the continuous amount of time in hours that a computer functions before breaking down.
- **Queueing & Logistics:**
  - Measuring the waiting time in minutes until the first customer or patient arrives at a facility.

Properties of Probability

- Sample Space, S
- The event set, E
- The probability measure, p

Properties of Probability: Examples

- **The experiment of tossing a coin:**
  - Sample Space, $S := \{H, T\}$
  - Event set of getting a Head, $E := \{H\}$
  - Probability measure of E: $p = \frac{1}{2}$

Probability measure	Description
$P(A) \geq 0$	Probability of an event A is 0 or greater
$P(S) = 1$	Probability of space S is 1
$0 \leq P(A) \leq 1$	Probability is always between 0 and 1.

Review of Set Algebra

Symbols	Description
$\emptyset$	The null set or empty set.
$A \subset B$	A is a subset of B.
$A \cup B$	the union of A and B, representing elements in either set.
$A \cap B$	The intersection of A and B, representing elements shared by both sets.
A'	The complement of A, representing everything outside of A.
Mutually Exclusive:	Events $A_1, A_2, \ldots, A_k$ are mutually exclusive if $A_i \cap A_j = \emptyset$ for $i \neq j$; they are disjoint sets.
Exhaustive Events:	Events $A_1, A_2, \ldots, A_k$ are exhaustive if their union forms the entire sample space, $A_1 \cup A_2 \cup \cdots \cup A_k = S$.

Properties of Set Operations

- **Commutative Laws:**
  - $A \cup B = B \cup A$.
  - $A \cap B = B \cap A$.
- **Associative Laws:**
  - $A \cup B \cup C = A \cup (B \cup C)$.
  - $A \cap B \cap C = A \cap (B \cap C)$.
- **Distributive Laws:**
  - $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.
  - $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.
- **De Morgan's Laws:**
  - $(A \cup B)' = A' \cap B'$.
  - $(A \cap B)' = A' \cup B'$.

Additional Properties of Probability

- Probability is a real-valued set function P that assigns a number $P(A)$ to each event A in the sample space S.

Properties of probability	Description
$P(A) \geq 0$	Probability of an event A is 0 or greater.
$P(S) = 1$	Probability of space S is 1.
$P(\emptyset) = 0$	Probability of an empty set $\emptyset$ is 0.
$0 \leq P(A) \leq 1$	Probability is always between 0 and 1.
$P(A') = 1 - P(A)$	Given event A. The probability of the complement of event A, i.e. A^c or A' is $1 - P(A)$
$P(A) = \sum_{i=1}^{\infty} P(A \cap B_i)$	
$P(A) \leq P(B)$	If $A \subset B$, then $P(A) \leq P(B)$
$P(B \cap A)$	Probability that elements exists in both events A and B
* $P(B \cap A') = P(B) - P(B \cap A)$	

Probability Calculation Example

- **Experiment:** Tossing a coin twice.
- **Sample Space:** $S := \{HH, HT, TH, TT\}$, where each outcome is equally likely with a probability of $1/4$. (Hint: *equally likely*: each elements in the set of equal chance of being selected hence each element has the probability of $1/4$)
  - Let $E := \{HH, HT\}$ be the event that the first toss is a head.
  - Let $F := \{HH, TH\}$ be the event that the second toss is a head.

  Find the probability of the union $P(E \cup F)$:

  Answer (step-by-step):

  1. $P(E \cup F) = P(E) + P(F) - P(E \cap F)$.
  2. $P(E \cup F) = 1/2 + 1/2 - P(\{HH\})$.
  3. $P(E \cup F) = 1 - 1/4 = 3/4$.
  4. Alternatively, $P(E \cup F) = P(\{HH, HT, TH\}) = 3/4$.

Probability Calculation Example

- **Experiment:** Tossing a fair dice.
- **Sample Space:** $S := \{1, 2, 3, 4, 5, 6\}$ where each outcome is equally likely
  - Let E be the event that consist of event numbers
  - Let F be the event that consist of numbers greater than 3

  Find the probability of the union $P(E \cup F)$:

  Answer (step-by-step):

  1. Probability for each element is $\frac{1}{2}$
  2. $E := \{2, 4, 6\}$, hence $P(E) = \frac{1}{2}$
  3. $E := \{4, 5, 6\}$, hence $P(F) = \frac{1}{2}$
  4. $(E \cap F) := \{4, 6\}$, hence $P(E \cap F) = \frac{1}{3}$
  5. $P(E \cup F) = P(E) + P(F) - P(E \cap F)$.
  6. $P(E \cup F) = 1 - 1/3 = \frac{2}{3}$.
  7. Alternatively, $P(E \cup F) = P(\{H2, 4, 5, 6\}) = \frac{4}{6} = \frac{2}{3}$.

Probability Calculation: In-class exercise (A practice to your quiz)

Task: To answer the following questions, you will need to pair up with a friend.

a) You are given 10 minutes. Try to answer the following questions.
b) After 10 minutes, switch your answer with your friend. 5 minutes, review your friends work and add any missing steps/ fix the calculation if needed.

1. If $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \cap B) = 0.3$, calculate the following:

a. $P(A \cup B)$
b. $P(A \cap B')$
c. $P(A' \cup B)$

[6 mark] 2. A fair die is rolled three times. Let
A_1 := Outcome is 1 or 2 on the first rolle,
A_2 := Outcome is 3 or 4 on the first rolle, and

Method of Enumeration & Multiplication Principle

- The method of enumeration is used to determine the **number of outcomes associated with the events of certain random experiments**.
- **Multiplication Principle:** If an experiment E_1 has n_1 outcomes and for each possible outcome, a second experiment E_2 has n_2 possible outcomes, the composite experiment E_1E_2 has n_1n_2 possible outcomes.
- **Example:** Tossing a coin (E_1 with 2 outcomes) and rolling a die (E_2 with 6 outcomes).
  - The total number of outcomes is $2 \times 6 = 12$.
  - The sample space represented in ordered pairs: $S = \{(H,1),(H,2),(H,3),(H,4),(H,5),(H,6),(T,1),(T,2),(T,3),(T,4),(T$
- If an experiment is repeated multiple times, such as tossing a coin ten times, the number of outcomes in the sample space is $2 \times 2 \times \cdots \times 2 = 2^{10}$.

Probability Distribution

- Experiment involves arrangement, selection and replacement.
- Suppose that we have 4 letter a,b,c and d, the number of possible four-letter code easily can be calculated if each letter can be repeated which is $4^4 = 256$
- If consider without repetition, then the number of possible four-letter code is the number of permutation of 4 letter which is $4! = 24$.

Permutations and Combinations (Sampling)

- **Arrangement and Replacement Example:** Given 4 letters a, b, c, and d.
  - With repetition: The number of possible four-letter codes is $4^4 = 256$.
  - Without repetition: The number of possible four-letter codes is a permutation of 4 letters, which is $4! = 24$.
- **Case 1 (Replacement, Ordered):** Sampling with replacement of an ordered sample of size r yields n^r arrangements, i.e. $r = n^r$.
- **Case 2 (No Replacement, Ordered - Permutation):** Sampling without replacement of r ordered items from n objects.
  - ${}_nP_r = \frac{n!}{(n-r)!}$. It must be true that $r \leq n$.
- **Case 3 (No Replacement, Unordered - Combination):** Sampling without replacement of r selected objects from n objects with unordered subsets.
  - ${}_nC_r = \frac{n!}{r!(n-r)!}$.

Permutation Calculation Example

1. How many four-letter code words are possiblle using the letters in **IOWA** if

a. The letters may not be repeated

b. The letters may be repeated

2. Suppose that two tennis players A and B are playing in a match. The first player that wins three sets will win the match. How many ways could this tennis match ends?